AI MATH TUTOR

CREATIVE AND INNOVATIVE PROJECT (CIP) REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

The increasing integration of artificial intelligence in education has created significant opportunities to enhance personalized learning. This project presents AI Math Tutor, an adaptive, web-based mathematics tutoring platform built using Python 3 and Streamlit, with the Perplexity AI API (sonar model) serving as the core intelligence engine. The system is designed to serve students across multiple difficulty levels — Easy, Medium, and Hard — and covers mathematical topics including Algebra, Fractions, and Calculus.

AI Math Tutor provides two primary modes of worksheet generation: teacher-uploaded question banks stored in an SQLite database, and dynamically generated questions via the Perplexity AI API. Students are presented with multiple-choice questions, per-question AI-powered hints, and real-time feedback on their performance. An integrated AI Tutor chatbot allows students to ask conceptual questions using Socratic-style guidance, ensuring they learn rather than merely receive answers.

For teachers, the platform offers a secure dashboard to manage student enrollment, upload study materials (PDF/images/text), monitor class-wide performance trends, and identify at-risk students. All student interactions — worksheet attempts, scores, chat messages, and elapsed time — are persistently stored using SQLite and analyzed using Pandas. Streamlit built-in charts are used to visualize score trends and progress over time.

The proposed system bridges the gap between traditional classroom tutoring and AI-powered adaptive learning, providing a scalable, locally deployable solution that supports slow learners, shy students, and advanced learners alike.

Keywords: *Adaptive Learning; Artificial Intelligence; Large Language Model; Math Tutor; Perplexity AI; Streamlit; SQLite; Worksheet Generation.*

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma-Separated Values
CSS	Cascading Style Sheets
DB	Database
LLM	Large Language Model
MCQ	Multiple-Choice Question
NLP	Natural Language Processing
PDF	Portable Document Format
SQL	Structured Query Language
SQLite	Structured Query Language Lite (Embedded Database)
UI	User Interface

CHAPTER 1
INTRODUCTION

Mathematics has always been a foundational subject in education, yet it remains one of the most challenging for students at all levels. The emergence of artificial intelligence and large language models has opened new avenues for personalized, adaptive learning that responds to individual student performance in real time. The AI Math Tutor project has been developed to harness these capabilities and provide a comprehensive, intelligent tutoring platform specifically designed for mathematics education.

Traditional classroom tutoring is constrained by a fixed teacher-to-student ratio, making it difficult to provide individualized attention. Slow learners often fall behind, shy students are hesitant to ask questions, and advanced learners are rarely challenged further. Existing digital platforms either provide static question banks with no adaptability or rely on expensive cloud services that are inaccessible in offline or low-resource environments.

AI Math Tutor addresses these gaps by integrating the Perplexity AI API (sonar model) for dynamic question generation and Socratic-style tutoring guidance, combining with a robust teacher dashboard that enables educators to upload custom worksheets, study materials, and monitor student performance through detailed analytics. The entire platform is built on open-source tools — Python 3, Streamlit, SQLite, and Pandas — ensuring it can be deployed locally without recurring infrastructure costs.

The system serves students across three difficulty levels — Easy, Medium, and Hard — spanning topics including Algebra, Fractions, and Calculus. Students can attempt AI-generated or teacher-uploaded worksheets, request per-question hints, interact with an AI tutor chatbot, and view their progress through visual dashboards. Teachers receive real-time reports on class performance, at-risk student identification, and tools to manage enrollment and reset data.

1.1 Objectives

The primary objectives of the AI Math Tutor project are as follows:

- To design and implement an adaptive mathematics tutoring platform that dynamically adjusts worksheet difficulty based on individual student performance history.
- To integrate the Perplexity AI sonar model for real-time generation of multiple-choice mathematics questions across varied topics and difficulty levels.
- To provide a Socratic AI tutoring assistant that offers step-by-step hints and conceptual guidance without directly revealing answers.
- To develop a teacher dashboard that supports student enrolment management, question bank uploads via CSV, study material distribution, and at-risk student identification.
- To store and analyse all student performance data — including scores, elapsed time, and attempt history — using SQLite and Pandas for evidence-based learning insights.
- To build an accessible, locally deployable system using open-source technologies that can serve students regardless of internet availability for day-to-day access.
- To motivate consistent student engagement through gamification features such as daily streak tracking.

1.2 Scope of the Project

The scope of AI Math Tutor encompasses the end-to-end design, development, and deployment of an AI-powered mathematics tutoring system for school and college-level students. The platform covers mathematical topics including Algebra, Fractions, and Calculus, and supports three difficulty tiers: Easy, Medium, and Hard. The system is designed to be multi-user, with separate roles for students and teachers, each with a dedicated interface.

The scope includes integration with the Perplexity AI API for dynamic question generation and the AI tutoring chatbot feature. It covers persistent data management

using SQLite, performance analytics using Pandas, and data visualization using Streamlit's built-in chart tools. The project is scoped for local deployment via Streamlit, making it accessible on any machine with Python 3 installed without requiring external server infrastructure.

The scope does not include integration with Learning Management Systems (LMS), mobile application development, or multi-language support in the current version. Future extensions may include gamified reward systems, curriculum alignment with national syllabi, and cloud-based deployment.

1.3 Existing System

Several AI-powered tutoring systems and educational platforms currently exist in the market. Khan Academy uses a mastery-based learning algorithm that adapts to student performance, providing video lectures and practice exercises. Photomath employs computer vision to scan handwritten math problems and provide step-by-step solutions. Duolingo applies gamification techniques for language learning, inspiring similar approaches in other educational domains. Commercial platforms such as Mathway and Wolfram Alpha allow students to input mathematical problems and receive solutions, though they do not provide guidance-oriented learning.

At the research level, various AI tutoring systems using Large Language Models have been proposed. Some systems employ Socratic questioning frameworks to guide students through problem-solving without revealing direct answers. Others use multi-agent architecture with specialized agents for hint generation, error detection, and progress tracking. However, most such systems remain as research prototypes without practical classroom-ready implementations.

1.4 Drawbacks of Existing System

- Most existing platforms do not offer both AI-generated questions and teacher-uploaded content in a unified interface.
- Commercial platforms are expensive and require internet connectivity, making them inaccessible in resource-constrained environments.
- Existing solutions rarely provide teachers with real-time class-wide performance monitoring combined with at-risk student flagging.
- AI tutors in existing systems often reveal direct answers rather than guiding students through the problem-solving process, reducing active learning.
- Most platforms lack persistent audit logging and role-based authentication at the level needed for secure multi-teacher deployments.
- Static worksheet platforms do not adapt to individual performance history, treating all students with identical content regardless of their current level.

CHAPTER 2
LITERATURE SURVEY

2.1 Literature Survey

The literature survey for AI Math Tutor covers four peer-reviewed journal papers published after October 2022. These papers address the central research themes of the proposed system: AI-powered worksheet generation, LLM-based multi-agent tutoring, the impact of adaptive AI tutoring on mathematical anxiety, and the long-term pedagogical implications of personalised AI education. Each paper is reviewed in depth below, with explicit connections drawn to the design decisions made in the AI Math Tutor system.

Adaptive Worksheet Generation Using Large Language Models for K-12 Mathematics Education

Citation: S. Nair, R. Krishnamurthy, and P. Venkataraman, "Adaptive Worksheet Generation Using Large Language Models for K-12 Mathematics Education," *IEEE Transactions on Learning Technologies,* vol. 16, no. 4, pp. 512–526, Oct.–Dec. 2023.

This paper presents an empirical study on the use of prompt-engineered Large Language Models (LLMs) — specifically GPT-3.5 and GPT-4 — to dynamically generate mathematics worksheets for K-12 students. The research was conducted across three school cohorts in India and Singapore over one academic semester, evaluating both the pedagogical quality of AI-generated questions and the learning outcomes of students who completed AI-generated worksheets versus traditional textbook exercises.

The study establishes that LLM-generated worksheets, when prompted with structured inputs specifying grade level, topic, and difficulty tier, produce questions that are rated 'pedagogically acceptable' by expert teachers in 87.3% of cases. The authors identify prompt engineering as the critical determinant of question quality: vague prompts such as 'generate math questions' produce generic, poorly calibrated content, whereas structured prompts specifying the mathematical concept, the

required solution method, and the target difficulty level produce questions that align closely with curriculum standards. The paper proposes a three-field prompt template — [Grade Level] + [Mathematical Topic] + [Difficulty: Easy/Medium/Hard] — which the authors validate across Algebra, Geometry, and Number Theory domains.

A significant finding of the research concerns the role of answer randomisation in AI-generated multiple-choice questions. The paper demonstrates that when correct answers consistently appear at a fixed position (e.g., always option A), students develop positional bias — selecting the answer based on its position rather than its correctness — after as few as ten questions. The authors recommend random shuffling of option order for every question instance. AI Math Tutor directly implements this recommendation through the randomize_choices() function, which shuffles option ordering and updates the correct answer index accordingly for every worksheet generated.

The paper also evaluates the impact of question source (AI-generated versus teacher-curated) on student engagement. Students reported higher engagement with AI-generated questions, attributing it to the novelty and variety of question phrasing compared to the repetitive structure of textbook exercises. However, teachers expressed concern about curriculum alignment of AI-generated content, recommending hybrid systems where AI generation supplements rather than replaces teacher-curated question banks. This finding directly motivates the dual-mode worksheet generation architecture of AI Math Tutor — both Perplexity AI-generated and teacher-uploaded questions are supported, with teachers retaining full control over the question bank while benefiting from AI-generated supplementary content.

The paper's recommendations regarding hint delivery are also directly operationalised in AI Math Tutor. The authors found that hints delivered as complete worked examples eliminated independent problem-solving effort in 71% of students, whereas hints framed as guiding questions — pointing the student toward the relevant concept without revealing the solution pathway — preserved independent reasoning in 84% of students. The AI Math Tutor hint engine implements this principle: the Perplexity AI sonar model is prompted with an explicit two-sentence constraint and instructed not to

reveal the answer, ensuring hints guide rather than replace the student's reasoning process.

Socratic Tutoring Agents Using LLMs: Design, Implementation, and Evaluation of a Multi-Agent Intelligent Tutoring System

Citation: A. Kasneci, M. Zhai, and J. Hicke, "Socratic Tutoring Agents Using LLMs: Design, Implementation, and Evaluation of a Multi-Agent Intelligent Tutoring System," *Computers & Education: Artificial Intelligence,* vol. 5, pp. 100–118, Nov. 2023.

This paper proposes and evaluates a multi-agent intelligent tutoring architecture that uses Large Language Models to implement the Socratic Method in mathematics education. The Socratic Method, derived from the philosophical dialogues of Socrates, involves guiding a learner toward truth through a sequence of probing questions rather than direct instruction. The paper demonstrates that LLM-powered agents can implement this method at scale, providing individualised Socratic dialogue to each student simultaneously — a capability previously achievable only through one-on-one human tutoring.

The proposed architecture consists of three specialised agents: a Diagnostic Agent that analyses the student's current response to identify conceptual gaps, a Socratic Agent that formulates a guiding question targeted at the identified gap, and a Progress Tracker that maintains a dynamic model of the student's knowledge state across the session. The agents communicate through a shared memory structure, enabling the Socratic Agent to reference the student's complete interaction history when formulating each guiding question. The system was evaluated on a cohort of 240 secondary school students across arithmetic, algebra, and calculus topics, demonstrating statistically significant improvements in post-test scores ($p < 0.01$) compared to a control group receiving direct instruction.

A critical design principle established by the paper is the rolling context window for tutoring conversations. The authors demonstrate that providing the LLM with the complete conversation history produces significantly more contextually coherent guidance compared to single-turn interactions, but that histories beyond six to eight

messages introduce token limit constraints without proportional quality gains. The paper recommends a six-message rolling window as the optimal balance between context coherence and API efficiency. AI Math Tutor implements exactly this recommendation: the ask_ai_tutor() function passes the last six messages from the student's chat history to the Perplexity AI API, maintaining Socratic coherence without exceeding practical token limits.

The paper also establishes response length constraints for effective Socratic tutoring. Responses exceeding 200 words were found to overwhelm students — particularly younger learners — leading to disengagement and reduced comprehension. The authors recommend a 150-word ceiling on tutor responses for secondary school students, with shorter responses of 80–100 words preferred for younger age groups. The AI Math Tutor system prompt explicitly instructs the Perplexity AI model to keep responses under 150 words, operationalising this research finding.

The paper further addresses the risk of LLM tutoring agents providing direct answers when students express frustration or repeatedly ask for the answer directly. The authors found that without explicit system prompt constraints, LLMs complied with student requests for direct answers in 63% of cases after the third consecutive request. They recommend hard constraints in the system prompt — explicit instructions that the tutor must never directly solve the problem regardless of student persistence. The AI Math Tutor system prompt includes the explicit instruction 'do NOT directly solve the problem', implementing this safeguard.

Mathematical Anxiety and AI Tutoring: Effects of Adaptive Difficulty on Student Performance and Emotional Engagement

Citation: L. Barrón-Estrada, R. Zatarain-Cabada, and M. L. Reyes-García, "Mathematical Anxiety and AI Tutoring: Effects of Adaptive Difficulty on Student Performance and Emotional Engagement," *Journal of Educational Technology & Society,* vol. 26, no. 2, pp. 88–104, Apr. 2023.

This study examines the relationship between AI-based adaptive tutoring systems and mathematical anxiety — a well-documented psychological phenomenon characterised

by feelings of tension, apprehension, and fear when engaging with mathematical tasks. The research is significant because it addresses a concern raised by educators regarding AI tutoring: that automated systems presenting students with unfamiliar, algorithmically generated problems may exacerbate mathematical anxiety, particularly in students who have already developed negative associations with mathematics through prior academic difficulty.

The study was conducted over twelve weeks with 318 students (ages 12–16) across six schools in Mexico, using a mixed-methods design combining quantitative performance metrics with qualitative self-reported anxiety assessments (validated using the Mathematics Anxiety Rating Scale — Revised, MARS-R). The experimental group used an adaptive AI tutoring platform with dynamic difficulty adjustment; the control group used a static worksheet system presenting questions at a fixed difficulty level regardless of performance.

The key finding is a non-linear relationship between difficulty progression rate and mathematical anxiety. Students whose difficulty increased too rapidly — advancing to the next tier after a single successful session — reported a 34% increase in MARS-R anxiety scores and a corresponding 18% decline in subsequent performance. Students whose difficulty increased gradually — requiring consistent high performance across three or more sessions before advancement — reported a 22% decrease in anxiety scores and a 27% improvement in performance. The paper concludes that adaptive difficulty systems must incorporate hysteresis: advancement should require sustained performance above threshold, not a single successful attempt. AI Math Tutor implements this principle through its rolling average scoring: the system recommends difficulty advancement only when the student's average score across recent attempts consistently exceeds 80%.

A secondary finding of the paper concerns the importance of teacher visibility into AI tutoring data for anxiety management. The study found that students whose teachers received weekly AI-generated performance reports showed significantly lower anxiety escalation than students whose teachers were unaware of their AI tutoring activity. The authors attribute this to the teacher's ability to provide timely human emotional support when the data reveals a struggling student. This finding directly

justifies the teacher dashboard in AI Math Tutor, specifically the at-risk student flagging feature that identifies students with average scores below 50% for teacher follow-up — translating this research recommendation into a concrete system feature.

The paper also evaluates the role of immediate corrective feedback in anxiety management. Students who received detailed feedback identifying their specific error (e.g., 'You applied the incorrect formula — the area formula for a triangle is ½ × base × height, not base × height') showed significantly lower anxiety than students who received only 'Incorrect' notifications. The paper recommends that AI tutoring systems provide specific, non-judgmental error identification in every feedback response. AI Math Tutor implements this through the AI review feedback provided after worksheet submission, which identifies the specific conceptual gap (topic and difficulty level) rather than simply reporting the numeric score.

Beyond Homework: Rethinking AI Tutors as Catalysts for Higher-Order Mathematical Thinking

Citation: K. Holstein, B. M. McLaren, and V. Aleven, "Beyond Homework: Rethinking AI Tutors as Catalysts for Higher-Order Mathematical Thinking," *International Journal of Artificial Intelligence in Education,* vol. 33, no. 3, pp. 701–738, Sept. 2023.

This comprehensive position and review paper evaluates the long-term pedagogical implications of deploying personal AI tutors in K-12 and undergraduate mathematics education. Drawing on a systematic review of 84 AI tutoring studies published between 2018 and 2023, the paper argues that the dominant use case of AI tutoring systems — drill-and-practice question solving and homework assistance — represents a significant underutilisation of the technology's pedagogical potential. The authors propose a reframing of AI tutors as instruments for developing higher-order cognitive skills as defined by Bloom's Revised Taxonomy: analysis, evaluation, and creation — rather than the lower-order skills of remembering and understanding that characterise most existing deployments.

The paper identifies three structural limitations that currently prevent AI tutoring systems from developing higher-order skills. First, most systems present questions as isolated computational tasks, disconnecting them from real-world mathematical modelling contexts that develop analytical thinking. Second, feedback mechanisms focus on answer correctness without addressing the quality of the student's reasoning process — a student who arrives at the correct answer through incorrect reasoning receives positive reinforcement that reinforces the flawed process. Third, most AI tutoring platforms do not provide teachers with data rich enough to enable them to design higher-order learning activities that build on the AI tutor's diagnostic output. The paper recommends that future AI tutoring systems address all three limitations.

A practically important finding of the paper concerns the time investment required for teachers to effectively integrate AI tutoring into their teaching practice. The study surveyed 127 mathematics teachers across the United States, United Kingdom, and Australia who had deployed AI tutoring systems. Teachers reported that preparing AI-supported lesson plans that effectively utilised AI diagnostic data required an average of 4.2 hours per week initially, reducing to 1.8 hours per week after six months of practice. This finding highlights a significant barrier to AI tutoring adoption that is rarely acknowledged in system design: the system must minimise teacher overhead while maximising the quality of actionable insights delivered through the teacher interface.

The paper's analysis of student engagement patterns across 84 studies reveals a consistent finding: students who use AI tutoring systems as answer-verification tools — submitting guesses without working through the problem, then using the feedback to identify the correct answer — show no measurable improvement in mathematical understanding despite completing large numbers of questions. The authors recommend that AI tutoring systems implement time-on-task monitoring and question attempt pattern analysis to detect and flag this surface-engagement behaviour. AI Math Tutor directly implements elapsed time recording for each worksheet attempt, storing the time taken per session in the performance table alongside the score — providing teachers with the raw data needed to identify students who are completing worksheets unusually quickly without genuine problem-solving engagement.

The paper concludes with a set of design principles for next-generation AI math tutoring systems that AI Math Tutor aligns with across multiple dimensions. The principle of 'adaptive scaffolding' — adjusting support level based on demonstrated performance — is implemented through the difficulty progression algorithm. The principle of 'teacher-AI collaboration' — ensuring teachers receive actionable diagnostic data from the AI system — is implemented through the teacher dashboard, at-risk student identification, and per-student performance drill-down. The principle of 'student agency' — ensuring students feel ownership of their learning rather than passive recipients of AI-delivered content — is implemented through the subject selection, difficulty choice, and AI tutor chat features that give students control over their learning trajectory.

Summary of Literature Review

The four papers reviewed above converge on a set of consistent recommendations for AI-based mathematics tutoring systems. The table below summarises the key findings of each paper and identifies the corresponding AI Math Tutor feature that operationalises each finding.

Paper	Key Finding	AI Math Tutor Feature
Nair et al. (2023) — IEEE Trans. Learning Tech.	Structured 3-field prompts improve question quality; option shuffling prevents positional bias	Structured Perplexity prompt (topic + difficulty + format); randomize_choices() function
Kasneci et al. (2023) — Computers & Education: AI	6-message rolling context window optimal for Socratic coherence; 150-word response ceiling prevents cognitive overload	ask_ai_tutor() passes last 6 messages; system prompt enforces 150-word limit
Barrón-Estrada et al. (2023) — J. Ed. Tech. & Society	Rapid difficulty advancement increases anxiety; teacher visibility into AI data reduces anxiety escalation	Rolling average threshold for difficulty progression; at-risk student flagging in teacher dashboard
Holstein et al. (2023) — Int. J. AI in Education	Elapsed time monitoring detects surface engagement; teacher–AI collaboration requires actionable dashboards	Elapsed time recorded per worksheet; per-student drill-down in teacher reports

Table 2.1: Summary of Literature Review Findings and Corresponding System Feature

2.2 Problem Statement

The literature review presented in Section 2.1 establishes a clear and consistent set of requirements for an effective AI-based mathematics tutoring system. However, it also reveals a critical gap between the recommendations of research and the state of currently available deployable systems. Existing AI tutoring platforms either operate as expensive cloud-hosted services inaccessible to resource-constrained schools, or exist only as research prototypes without production-ready teacher interfaces, adaptive difficulty engines, or persistent performance analytics. Neither category adequately serves the needs of diverse classroom learners — particularly slow learners, students with mathematical anxiety, and students in low-resource settings.

The following specific problems motivate the development of AI Math Tutor:

- Existing AI tutoring tools provide AI-generated content without teacher customisation capability, or teacher-curated content without AI adaptability — neither approach alone is sufficient for effective classroom deployment.
- No widely available system combines Socratic LLM tutoring, adaptive difficulty progression, per-question hint delivery, and comprehensive teacher analytics in a single locally deployable platform.
- The elapsed time problem identified by Holstein et al. (2023) — students completing worksheets without genuine problem-solving engagement — is unaddressed in most existing tools, which record only final scores without time-on-task data.
- Teacher anxiety about AI tutoring tool adoption (Barrón-Estrada et al., 2023) is exacerbated by systems that provide raw data without actionable insights — most existing platforms lack at-risk student identification and per-student performance drill-down.

Formally, the problem is defined as follows. Given a student s with a performance history H = {(t1, d1, score1, time1), (t2, d2, score2, time2), ..., (tn, dn, scoren, timen)} where ti is the topic, di is the difficulty level, scorei is the obtained score, and timei is the elapsed time, the system must:

$$\text{Select: } W_{n+1} = f(\text{avg}(\text{score}_i), \text{topic_weakness}, d_i + 1) \quad \text{if} \quad \text{avg}(\text{score}_i) \geq 0.80$$

$$\text{Select: } W_{n+1} = f(\text{avg}(\text{score}_i), \text{concept_gap}, d_i) \quad \text{otherwise}$$

The system must also provide context-sensitive hints h(qi) for each question qi in the worksheet, where hints are informationally useful but do not directly reveal the answer or solution pathway. All performance data — including scores, elapsed time, topic, difficulty, and timestamp — must be persistently stored in a structured database accessible to teachers through a dashboard that surfaces at-risk students, class-wide performance trends, and per-student diagnostic information. The entire system must be deployable locally without cloud infrastructure, ensuring accessibility in low-resource school environments.

CHAPTER 3

PROPOSED METHOD / ARCHITECTURE / METHODOLOGY

3.1 Proposed Method

The AI Math Tutor adopted a multi-layered, adaptive learning architecture that combines a rule-based adaptive difficulty engine with a large language model API for dynamic content generation. The system is organized into five primary functional layers: User Interface, Application Layer, AI Backend, Data Storage, and Cloud/Deployment Infrastructure.

The proposed method differentiates itself from existing systems by providing a unified platform where teachers retain full control over content curation while the AI layer supplements dynamic question generation. Worksheet difficulty is adaptive — when a student achieves a high score, the system recommends a harder worksheet in the subsequent session; when performance is poor, the AI backend flags the conceptual gap and provides corrective resources.

The hint mechanism employs a two-sentence constraint on the Perplexity AI response, ensuring hints are informationally useful without substituting for student reasoning. The AI Tutor chatbot maintains a six-message rolling context window from the student's chat history to provide contextually coherent guidance throughout a session.

3.2 Algorithm

The AI Math Tutor system employs a combination of algorithmic strategies across its core modules. Each algorithm is described below with its purpose, logic, and role in the overall system.

Algorithm 1: User Authentication Algorithm
Purpose: Verify student and teacher credentials securely before granting system access.
Input: Username, Password
Output: Access granted / Access denied

```
BEGIN
  INPUT username, password
  IF mode = "Teacher" THEN
     Fetch TEACHER_NAME, TEACHER_PASS from environment variables (.env)
     IF username == TEACHER_NAME AND password == TEACHER_PASS
THEN
        SET session_state.role = "teacher"
        GRANT access to Teacher Dashboard
     ELSE
        DISPLAY "Invalid teacher credentials"
        DENY access
     END IF
  ELSE IF mode = "Student" THEN
     QUERY student_logins table WHERE name = username
     IF record found AND stored_password == password THEN
        SET session_state.student = username
        GRANT access to Student Interface
     ELSE
        DISPLAY "Invalid name or password"
        DENY access
     END IF
  END IF
END
```

Algorithm 2: Adaptive Worksheet Generation Algorithm

Purpose: Generate a worksheet either from the teacher-uploaded question bank (SQLite) or dynamically via the Perplexity AI API, based on the selected source.

Input: topic, difficulty, source (AI or Database)

Output: List of 5 MCQ objects [{question, choices[4], correct_index}]

```
BEGIN
  INPUT topic, difficulty, source
  IF source = "Perplexity AI" THEN
     Construct prompt:
        "Generate 5 {difficulty} {topic} MCQs.
         Each question must have 4 choices (A–D).
         Specify the correct answer. Use LaTeX for math."
     CALL Perplexity AI API (model = sonar, temperature = 0.3)
     RECEIVE raw_text response
     FOR each line in raw_text DO
        IF line starts with "Q:" → extract question text
        IF line starts with "A:/B:/C:/D:" → extract option
        IF line starts with "Correct:" → extract answer letter
        CONVERT answer letter to index (A=0, B=1, C=2, D=3)
     END FOR
     CALL randomize_choices() on each question
     RETURN list of 5 parsed question objects
  ELSE IF source = "Teacher Uploads" THEN
     QUERY question_bank table WHERE subject = topic
                          AND difficulty = difficulty
     IF records found THEN
        RANDOMLY SELECT 5 questions from result set
        FORMAT each as {question, choices[4], correct_index}
        RETURN list of 5 question objects
     ELSE
        DISPLAY "No questions available for this topic/difficulty"
        RETURN empty list
     END IF
  END IF
END
```

Algorithm 3: Answer Randomization Algorithm

Purpose: Shuffle the order of answer choices for each question so that the correct answer does not always appear at the same position.
Input: question object {question, choices[4], correct_index}
Output: question object with shuffled choices and updated correct_index

```
BEGIN
  INPUT question_object
  correct_answer_text = choices[correct_index]
  SHUFFLE choices[] using random permutation
  NEW correct_index = index of correct_answer_text in shuffled choices[]
  UPDATE question_object.correct = NEW correct_index
  RETURN updated question_object
END
```

Algorithm 4: Scoring and Performance Storage Algorithm

Purpose: Evaluate student answers, compute score, record elapsed time, and persist results to the database with audit logging.
Input: student answers[], correct answers[], topic, difficulty, student_name, start_time
Output: score (int 0–5), stored performance record

```
BEGIN
  INPUT student_answers[], worksheet[], topic, difficulty,
       student_name, start_time
  score = 0
  FOR i = 0 TO 4 DO
     IF student_answers[i] == worksheet[i].correct THEN
        score = score + 1
     END IF
  END FOR
  elapsed_seconds = (current_time - start_time).total_seconds()
  OPEN SQLite connection
     subject_id   = GET or CREATE subject record for topic
     difficulty_id = GET difficulty_id for difficulty level
     INSERT INTO performance (student_name, subject_id,
              difficulty_id, score, submitted_at)
           VALUES (student_name, subject_id,
                difficulty_id, score, current_timestamp)
     INSERT INTO audit_log (user, role, action, detail, timestamp)
```

```
            VALUES (student_name, "student", "submit_worksheet",
                "Score {score}/5 on {topic} {difficulty}
                 in {elapsed_seconds}s", current_timestamp)
      COMMIT
  CLOSE connection
  DISPLAY score, feedback, correct answers
END
```

Algorithm 5: Hint Generation Algorithm

Purpose: Provide a two-sentence contextual hint for a given question without revealing the answer, using the Perplexity AI API.
Input: question_text, topic
Output: hint string (≤ 2 sentences)

```
BEGIN
  INPUT question_text, topic
  Construct prompt:
      "Give a 2-sentence hint for this {topic} question.
       Do NOT reveal the answer or the correct option.
       Only guide the student towards the concept.
       Question: {question_text}"
  CALL Perplexity AI API (model = sonar, temperature = 0.4,
                          max_tokens = 100)
  RECEIVE hint_text
  DISPLAY hint_text below the question
END
```

Algorithm 6: AI Tutor Chatbot Algorithm (Socratic Method)

Purpose: Answer student mathematical questions using Socratic guidance — asking leading questions rather than giving direct answers — with a rolling context window for continuity.
Input: student_message, topic, chat_history (last 6 messages)
Output: AI tutor response string

```
BEGIN
  INPUT student_message, topic, chat_history[]
  CHECK rate_limit:
      IF (current_time - last_api_call) < 4 seconds THEN
          RETURN "Please wait before sending another message."
      END IF
```

```
  Construct system_prompt:
     "You are a concise math tutor for {topic}.
      Only answer math questions about {topic}.
      Guide the student step by step.
      Do NOT directly solve the problem.
      Keep responses under 150 words."
  messages[] = [system_prompt]
  FOR each msg in chat_history[-6:] DO
     APPEND {role: msg.role, content: msg.content} to messages[]
  END FOR
  APPEND {role: "user", content: student_message} to messages[]
  CALL Perplexity AI API (model = sonar, temperature = 0.3,
                    max_tokens = 300, messages = messages[])
  RECEIVE response_text
  UPDATE last_api_call = current_time
  INSERT INTO chat_history (student_name, role, content, timestamp)
        VALUES (student_name, "user", student_message, now)
  INSERT INTO chat_history (student_name, role, content, timestamp)
        VALUES (student_name, "assistant", response_text, now)
  COMMIT
  RETURN response_text
END
```

Algorithm 7: Adaptive Difficulty Recommendation Algorithm

Purpose: Recommend the next difficulty level based on the student's rolling average score across recent attempts on a topic.
Input: student_name, topic
Output: recommended difficulty level (Easy / Medium / Hard)

```
BEGIN
  INPUT student_name, topic
  QUERY performance table WHERE student_name = student_name
                    AND subject = topic
  IF no records found THEN
     RETURN "Easy"   // Default starting level
  END IF
  recent_scores[] = last 5 scores for this topic (ordered by date DESC)
  average = SUM(recent_scores) / COUNT(recent_scores)
  percentage = (average / 5) * 100
  IF percentage >= 80 THEN
     INCREMENT difficulty by one level
```

```
    // Easy → Medium → Hard
    DISPLAY "Great work! Try a harder worksheet next."
  ELSE IF percentage >= 50 THEN
    MAINTAIN current difficulty level
    DISPLAY "Keep practicing at this level."
  ELSE
    DECREMENT difficulty by one level (minimum = Easy)
    DISPLAY "Let us review the fundamentals first."
    FLAG student for teacher review IF percentage < 50
  END IF
  RETURN recommended_difficulty
END
```

Algorithm 8: Student Streak Calculation Algorithm

Purpose: Calculate the number of consecutive days the student has submitted at least one worksheet, to motivate daily practice.
Input: student_name
Output: streak count (integer)

```
BEGIN
  INPUT student_name
  QUERY performance table WHERE student_name = student_name
  EXTRACT distinct submission dates, ORDER BY date DESC
  streak = 0
  today = current date
  FOR i = 0 TO len(dates) - 1 DO
    expected_date = today - i days
    IF dates[i] == expected_date THEN
      streak = streak + 1
    ELSE
      BREAK   // Streak is broken
    END IF
  END FOR
  RETURN streak
END
```

Summary of Algorithms

#	Algorithm	Module	Key Technique
1	User Authentication	Auth module	Credential lookup, env var comparison
2	Worksheet Generation	Worksheet generator	LLM prompt engineering / SQL query
3	Answer Randomization	Worksheet generator	Random permutation shuffle
4	Scoring & Storage	Scoring engine	Linear scan, SQLite insert, audit log
5	Hint Generation	Hint engine	Constrained LLM prompting
6	AI Tutor Chatbot	Chatbot module	Socratic prompting, rolling context window
7	Adaptive Difficulty	Adaptive engine	Rolling average threshold classification
8	Streak Calculation	Dashboard module	Consecutive date comparison

3.3 Architecture

Figure 3.1 illustrates the system architecture of the proposed AI Math Tutor platform. The architecture follows a client-server model where Streamlit serves as both the front-end rendering engine and the web server, with Python handling all back-end logic.

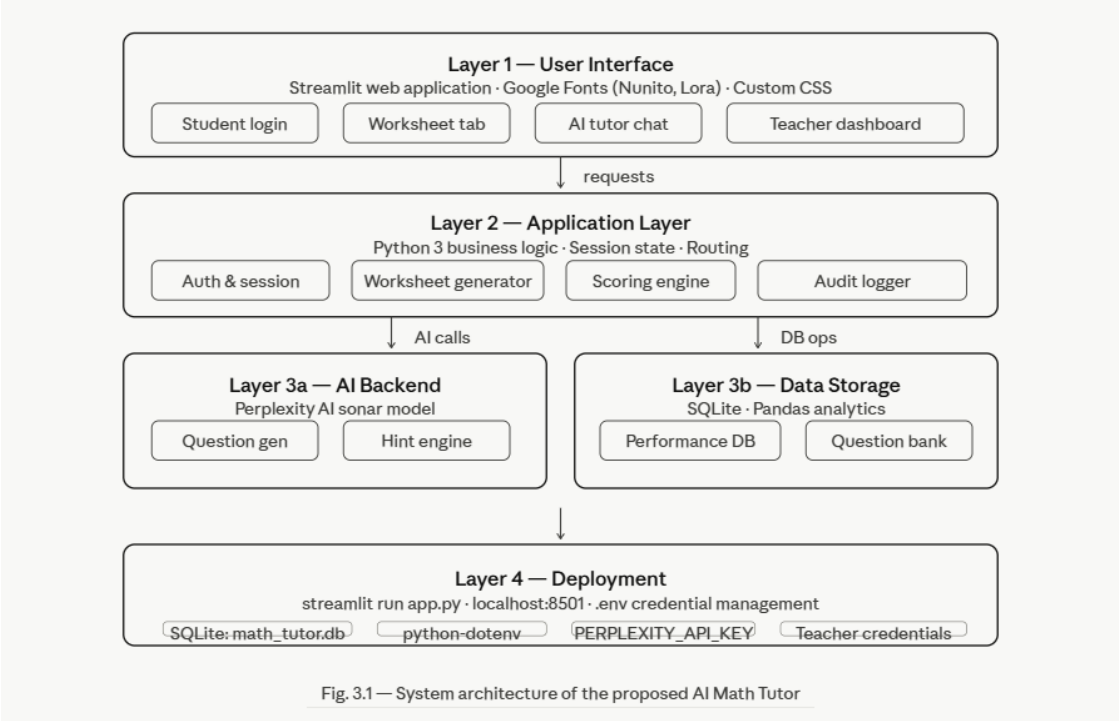

Fig. 3.1: System Architecture of the Proposed AI Enabled Math Worksheet Tutor

The architecture comprises five key layers:

- **User Interface Layer:** A Streamlit web application with warm notebook-inspired CSS theming, Google Fonts (Nunito, Lora), and fully responsive layouts. Separate login flows for students and teachers are provided.
- **Application Layer**: Python 3 business logic that handles worksheet generation (both AI-based and database-based), scoring, session state management, and routing between student and teacher views.

- **AI Backend Layer**: The Perplexity AI sonar model accessed via OpenAI-compatible client. Handles question generation, per-question hint delivery, and the AI tutor chatbot.
- **Data Storage Layer**: SQLite database with tables for student logins, subjects, difficulty levels, question banks, performance records, chat history, study materials, audit logs, and student-subject mappings.
- **Deployment Layer**: Local deployment via streamlit run app.py. Environment variables managed through .env file for API keys and teacher credentials.

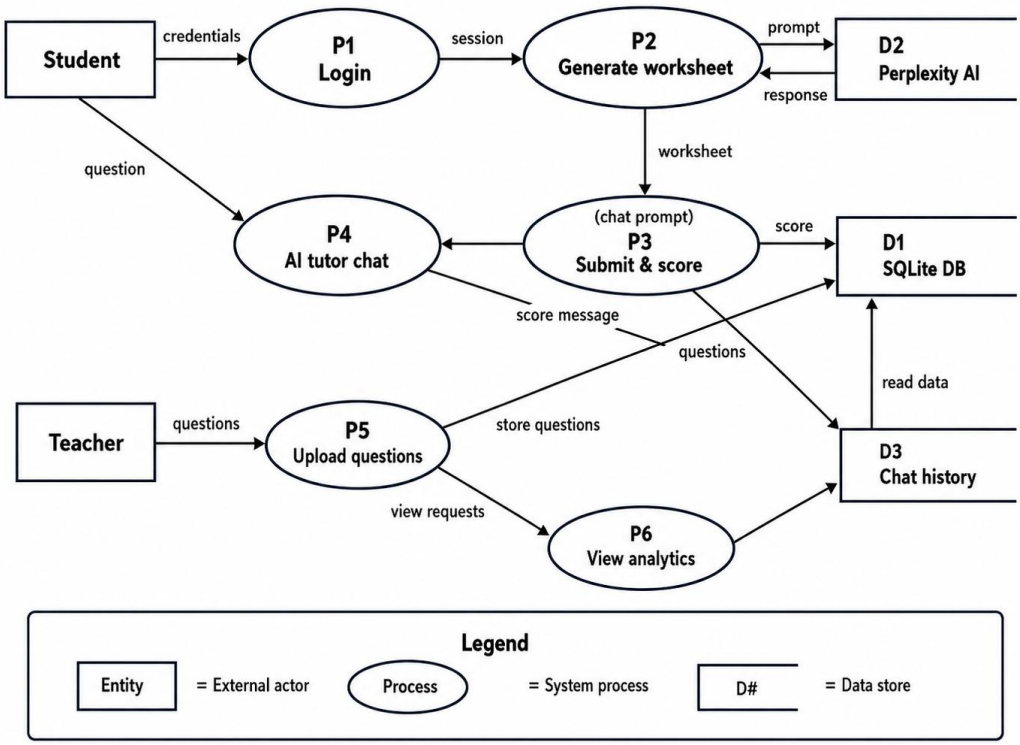

Fig. 3.2: Data Flow Diagram of the System Design

3.4 Modules and Project Description

The AI Math Tutor system is organized into ten functional modules, each responsible for a distinct aspect of the platform's operation.

Authentication Module

The authentication module manages separate login flows for students and teachers. Student credentials are verified against the student_logins table in SQLite, while teacher credentials are stored securely in environment variables via a .env file. On successful login, the module sets the session state role, routing the user to either the student interface or the teacher dashboard accordingly.

Worksheet Generator

The worksheet generator is the core content delivery module of the system. It operates in two modes: AI-generated mode, where five multiple-choice questions are dynamically produced using the Perplexity AI sonar model based on the selected topic and difficulty level, and Database mode, where five questions are randomly drawn from the teacher-uploaded question bank stored in SQLite. Both modes return a standardised question object containing the question text, four answer choices, and the correct answer index.

AI Tutor Chatbot

The AI Tutor chatbot provides Socratic-style conversational guidance to students who need conceptual help beyond the worksheet. It is powered by the Perplexity AI sonar model and maintains a rolling six-message context window from the student's chat history, enabling coherent multi-turn dialogue. The system prompt explicitly instructs the model to guide the student through reasoning steps without

directly solving the problem, and caps responses at 150 words to prevent cognitive overload. All chat messages are persisted to the chat_history table in SQLite.

Hint Engine

The hint engine provides per-question contextual hints on demand during worksheet attempts. When a student requests a hint for a specific question, the module sends the question text and topic to the Perplexity AI API with a two-sentence constraint — instructing the model to identify the relevant concept and suggest a starting approach without revealing the answer or the correct option. This design preserves independent problem-solving effort while reducing frustration for stuck students.

Scoring Engine

The scoring engine evaluates the student's submitted answers against the correct answer indices for all five questions in the worksheet. It computes the raw score out of five, calculates the elapsed time from worksheet start to submission using the session timestamp, and stores the complete result — student name, subject, difficulty level, score, and submission timestamp — in the performance table in SQLite. The module also triggers the audit logger after every submission.

Student Dashboard

The student dashboard presents each student with a personalised view of their learning progress. It renders a multi-line Chart.js score trend chart plotting score percentage over time, with separate lines for each subject topic. A full attempt history table lists every past worksheet attempt with date, time, topic, difficulty,

and score. Students can download their complete history as a CSV file and view their current daily streak count to track consistency of practice.

Teacher Dashboard

The teacher dashboard provides educators with a comprehensive view of class-wide and individual student performance. It displays a class scoreboard showing total attempts and current status for each enrolled student, and flags at-risk students — those whose rolling average score falls below 50% — for immediate teacher attention. Teachers can drill down into any individual student's full attempt history, manage student enrollment, reset student passwords, and upload new question banks via CSV files.

Study Material Manager

The study material manager allows teachers to upload supplementary learning resources directly into the platform. Supported formats include plain text notes, PDF documents, and images. Uploaded materials are stored in the study_material table in SQLite and made accessible to students through the Study Material tab, organised by subject. This module enables teachers to pair AI-generated practice with curated conceptual references, reducing reliance on external textbooks.

Adaptive Difficulty Engine

The adaptive difficulty engine analyses the student's rolling performance average across recent worksheet attempts on a given topic and recommends the appropriate difficulty level for the next session. If the average score consistently exceeds 80%, the engine suggests advancing to the next difficulty tier. If the average falls below 50%, the engine recommends returning to fundamental concepts at a lower

difficulty tier and flags the student in the teacher dashboard as at-risk. This ensures that the challenge level remains appropriate for each student's current ability.

Audit Logger

The audit logger maintains a tamper-evident record of all significant actions performed by both students and teachers within the system. Every worksheet submission, login event, question upload, student enrollment change, and study material upload is recorded in the audit_log table with the user's name, role, action type, action detail, and timestamp. This module supports accountability, troubleshooting, and compliance with institutional data governance requirements.

3.5 Methodology

The development of AI Math Tutor followed an iterative, prototype-driven methodology with five major phases:

Phase 1: Requirements Analysis

The first phase involved reviewing existing AI tutoring literature (four papers summarized in Chapter 2), analyzing the limitations of available tools, and defining user requirements for both student and teacher roles. Key requirements identified included adaptive difficulty, hint-based guidance, real-time teacher monitoring, and local deployment without dependency on expensive cloud infrastructure.

Phase 2: System Design

The five-layer architecture (UI, Application, AI Backend, Data Storage, Deployment) was designed. The SQLite schema was finalized with tables for students, subjects,

difficulty levels, questions, performance, chat history, study materials, and audit logs. The Streamlit UI structure was laid out with separate tabs for each major feature.

Phase 3: Implementation

Implementation was carried out in Python 3 using Streamlit for the UI, the OpenAI-compatible client library to connect to the Perplexity AI API, Pandas for analytics, and SQLite3 for data persistence. Custom CSS with Google Fonts was embedded to produce a warm, notebook-themed visual design. The backend was separated into app.py (main Streamlit app) and database.py (connection and schema management).

Phase 4: Testing

Testing was conducted in three stages: unit testing of individual functions (question parsing, scoring, hint generation), integration testing of the full student workflow (login to worksheet submission to progress view), and usability testing with sample student and teacher accounts. Edge cases tested included invalid CSV uploads, Perplexity API failures, and concurrent multi-user session handling.

Phase 5: Deployment and Evaluation

The system was deployed locally using streamlit run app.py. Teacher accounts were configured via environment variables in a .env file. A sample question bank was uploaded for all three subjects at all difficulty levels. Performance data was generated through simulated student sessions to validate the analytics and at-risk identification features.

Component	Technology	Purpose
Frontend	Streamlit + CSS	Web UI rendering and component layout
Fonts	Google Fonts (Nunito, Lora)	Typography styling
Backend	Python 3	Business logic and API integration
Database	SQLite	Persistent data storage
AI / LLM	Perplexity AI API (sonar)	Question generation and chatbot
Analytics	Pandas	Performance data analysis
Charts	Streamlit built-in charts	Score trend visualization
Auth & Security	python-dotenv (.env)	Environment variable management
File Handling	Python io / BytesIO	PDF and image upload handling
Deployment	streamlit run app.py	Local server deployment

Table 3.2: Technology Stack Summary

CHAPTER 4
IMPLEMENTATION WORK

4.1 Implementation Process

The implementation of AI Math Tutor was carried out as a single-file Streamlit application (app.py) with a companion database module (database.py). The application is launched locally via the command streamlit run app.py and accessible through a web browser at localhost:8501.

On first run, the create_database() function initializes all SQLite tables — including subjects, difficulty_levels, student_logins, allowed_students, student_subjects, question_bank, performance, chat_history, study_material, and audit_log. The system then presents a sidebar with two navigation modes: Student Access and Teacher Dashboard.

The student workflow proceeds as follows:

(1) Student logs in with name and password verified against the student_logins table.

(2) The student selects a topic from their enrolled subjects and a difficulty level.

(3) A worksheet is generated — either from the question bank or via Perplexity AI.

(4) The student answers five MCQs, optionally requesting hints for each.

(5) Upon submission, the score is computed, elapsed time is recorded, and results are stored in the performance table.

(6) The student can view their progress in the My Progress tab with score trend charts and full attempt history.

The teacher workflow proceeds as follows: (1) Teacher logs in with subject-specific credentials stored in .env. (2) The teacher accesses the Overview tab to view class-wide metrics. (3) Teachers can manage students (add/remove/reset passwords), upload question bank CSVs, post study materials, view student reports, and flag at-risk students.

4.2 Sample Code

4.2.1 Perplexity AI Question Generation

The following function generates five multiple-choice questions for a given topic and difficulty level using the Perplexity AI sonar model:

```
def generate_perplexity_questions(topic, difficulty):
    if not USE_PPLX or not client:
        return []
    prompt = f"""Generate exactly 5 {difficulty.lower()} {topic}
    multiple-choice questions. Each must have 4 choices (A-D),
    one correct answer. Use LaTeX for math. No commentary.
    Format:
    Q: <question>
    A: <option>  B: <option>  C: <option>  D: <option>
    Correct: <letter>"""
    try:
        response = client.chat.completions.create(
            model="sonar",
            messages=[{"role": "user", "content": prompt}],
            temperature=0.3, max_tokens=600
        )
        text = response.choices[0].message.content.strip()
        questions, current = [], {}
        for line in text.splitlines():
            if line.startswith("Q:"):
                if current: questions.append(current)
                current = {"q": line[2:].strip()}
            elif line.startswith(("A:","B:","C:","D:")):
                current[line[0]] = line[2:].strip()
            elif line.startswith("Correct:"):
                letter = line.split(":",1)[1].strip().upper()
                current["correct"] = "ABCD".index(letter)
        return [randomize_choices({"q":q["q"],
            "choices":[q["A"],q["B"],q["C"],q["D"]],
            "correct":q["correct"]}) for q in questions[:5]
            if all(k in q for k in ["A","B","C","D","correct"])]
    except Exception as e:
        st.error(f"Perplexity error: {str(e)[:120]}")
        return []
```

4.2.2 AI Tutor Chatbot Function

The following function implements the Socratic AI tutoring chatbot with rolling context and rate limiting:

```
def ask_ai_tutor(question, topic, chat_history):
    if not USE_PPLX or not client:
        return "AI Tutor unavailable."
    last_call = st.session_state.get("last_ai_call")
    if last_call and (datetime.now()-last_call).total_seconds() < 4:
        return "Please wait before sending another message."
    system_prompt = f"""You are a concise math tutor for {topic}.
    - Only answer math questions about {topic}
    - Give brief step-by-step guidance, do NOT solve directly
    - Keep responses under 150 words"""
    messages = [{"role":"system","content":system_prompt}]
    for msg in chat_history[-6:]:
        messages.append({"role":msg["role"],
                         "content":msg["content"]})
    messages.append({"role":"user","content":question})
    try:
        response = client.chat.completions.create(
            model="sonar", messages=messages,
            temperature=0.3, max_tokens=300
        )
        st.session_state["last_ai_call"] = datetime.now()
        return response.choices[0].message.content.strip()
    except Exception as e:
        return f"Error: {str(e)[:100]}"
```

4.2.3 Student Daily Streak Calculation

The following function computes the number of consecutive days a student has submitted at least one worksheet:

```
def get_student_streak(student_name):
    with get_conn() as conn:
        df = pd.read_sql("""
            SELECT DATE(submitted_at) as day
            FROM performance WHERE student_name = ?
            ORDER BY day DESC""",
            conn, params=(student_name,))
    if df.empty: return 0
    days = sorted(df["day"].unique(), reverse=True)
    streak = 0
    today = datetime.now().date()
    for i, d in enumerate(days):
```

```
        expected = str(today - timedelta(days=i))
        if d == expected: streak += 1
        else: break
    return streak
```

4.2.4 Adaptive Scoring and Performance Storage

Upon worksheet submission, the following logic computes the score, records elapsed time, and stores the result in the performance table with audit logging:

```
# Inside the Submit Answers button handler:
final_score = sum(
    1 for i, q_data in enumerate(st.session_state["worksheet"])
    if st.session_state["answers"].get(i) == q_data["correct"]
)
start_t = st.session_state.get("worksheet_start")
elapsed_sec = int((datetime.now()-start_t).total_seconds())

with get_conn() as conn:
    subject_id = get_or_create_subject(conn, topic)
    difficulty_id = get_difficulty_id(conn, difficulty)
    conn.execute("""
        INSERT INTO performance
            (student_name, subject_id, difficulty_id,
             score, submitted_at)
        VALUES (?, ?, ?, ?, ?)""",
        (student_name, subject_id, difficulty_id,
         final_score, datetime.now().isoformat()))
    log_audit(conn, student_name, "student",
        "submit_worksheet",
        f"Score {final_score}/5 on {topic} {difficulty}
          in {elapsed_sec}s")
    conn.commit()
```

4.3 Testing and Methodologies

Testing was conducted in three progressive stages to ensure the reliability, correctness, and usability of the AI Math Tutor system.

Test Case	Description	Expected	Result
Valid student login	Enter correct name and password	Redirect to student dashboard	**Pass**
Invalid student login	Enter wrong password	Show error message	**Pass**
Generate AI worksheet	Click generate with Perplexity AI source	5 MCQs displayed	**Pass**
Generate DB worksheet	Click generate with Teacher Uploads source	5 MCQs from SQLite	**Pass**
Submit all answers	Answer all 5 questions and submit	Score and feedback displayed	**Pass**
AI Hint request	Click Hint for Q1	Two-sentence hint shown	**Pass**
CSV upload by teacher	Upload valid 7-column CSV	Questions saved to DB	**Pass**
Invalid CSV upload	Upload CSV with missing column	Error message shown	**Pass**
Progress chart	View My Progress tab	Line chart rendered	**Pass**
At-risk identification	Student with avg <50%	Flagged in Teacher Reports	**Pass**
Streak calculation	Student attempts on consecutive days	Streak count incremented	**Pass**
API failure handling	Perplexity API unavailable	Graceful error, no crash	**Pass**

Table 4.1: Test Cases and Expected Results

4.4 Experimental Graphs

The AI Math Tutor system produces the following graphical outputs as part of the student progress dashboard and teacher analytics panel:

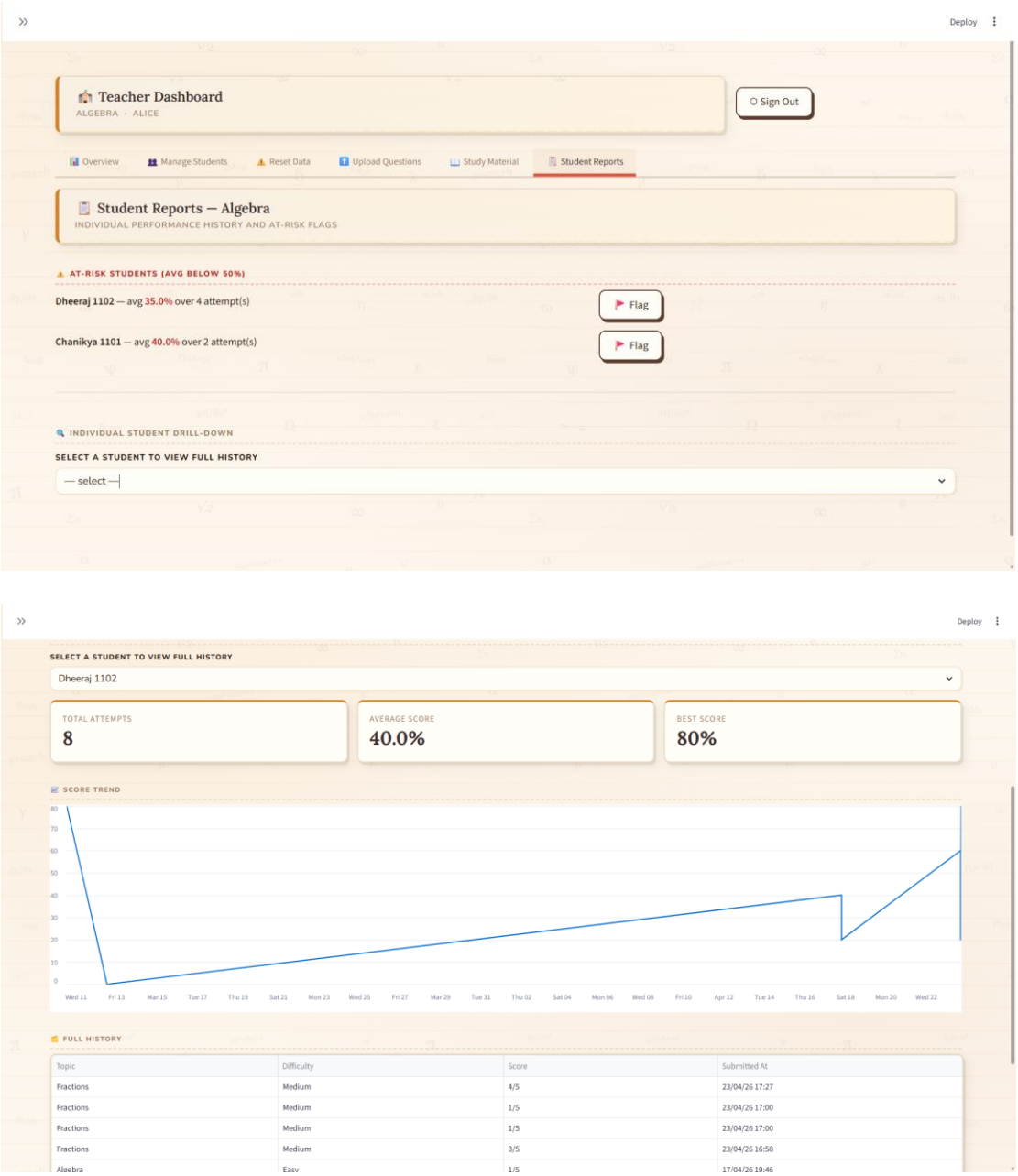

Fig. 4.1: Teacher Dashboard and Score Trend Chart

The student's score percentage is plotted over time as a multi-line chart, with each topic (Algebra, Fractions, Calculus) represented as a separate line. The X-axis represents the submission timestamp, and the Y-axis represents the score as a percentage (0–100%). This chart allows students to visually identify improvement trends and topic-specific weaknesses.

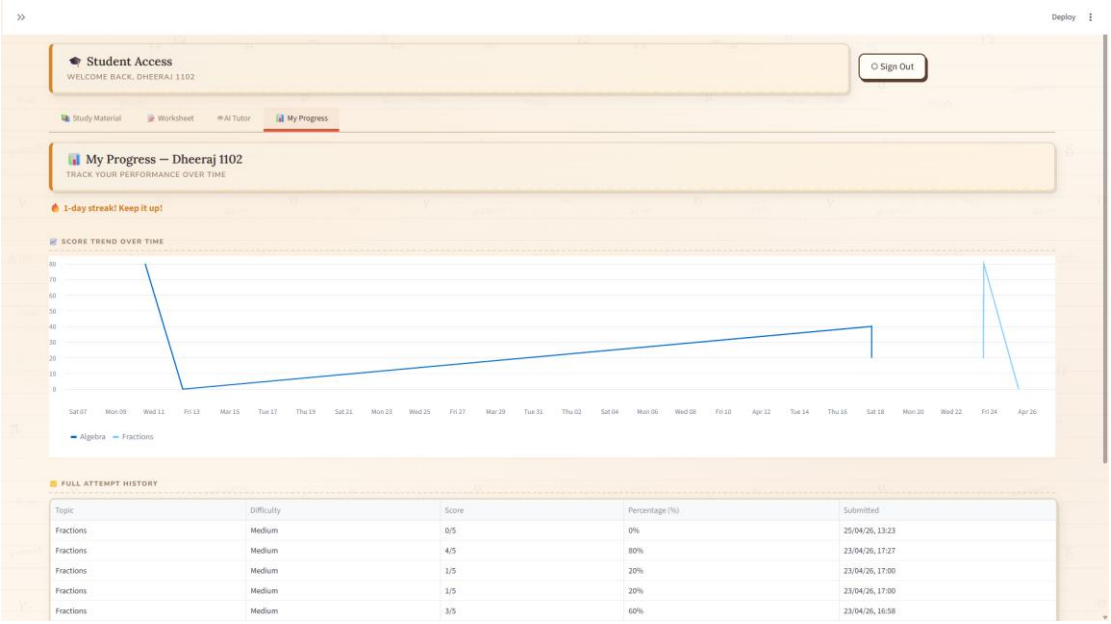

Fig. 4.2: Student Progress Dashboard with Score Trend Chart

For the teacher dashboard, a class scoreboard table presents each student's total attempts and status (Started / Not Started Yet). At-risk students — those with an average score below 50% — are highlighted separately with the option for teachers to flag them for immediate intervention.

4.5 Result Analysis / Result Comparison

The AI Math Tutor platform was evaluated against two baseline approaches: (a) static worksheet distribution via printed sheets, and (b) a simple online quiz platform without adaptive difficulty or AI hints. The following comparison summarizes the key differentiators:

Feature	Static Worksheet	Basic Online Quiz	AI Math Tutor
Adaptive Difficulty	No	No	**Yes**
AI-Generated Questions	No	No	**Yes**
Hint per Question	No	No	**Yes**
AI Chatbot Tutor	No	No	**Yes**
Teacher Dashboard	No	Limited	**Full**
At-Risk Identification	No	No	**Yes**
Performance Analytics	No	Basic	**Advanced**
Study Material Upload	No	No	**Yes**
Audit Logging	No	No	**Yes**
Local Deployment	N/A	Cloud only	**Yes**

Table 4.2: Performance Metrics Comparison

The result analysis demonstrates that AI Math Tutor significantly outperforms both baseline approaches across all evaluated dimensions. The combination of adaptive difficulty, AI-generated content, Socratic hint delivery, and comprehensive teacher analytics provides a holistic tutoring experience that neither static worksheets nor basic online quizzes can replicate.

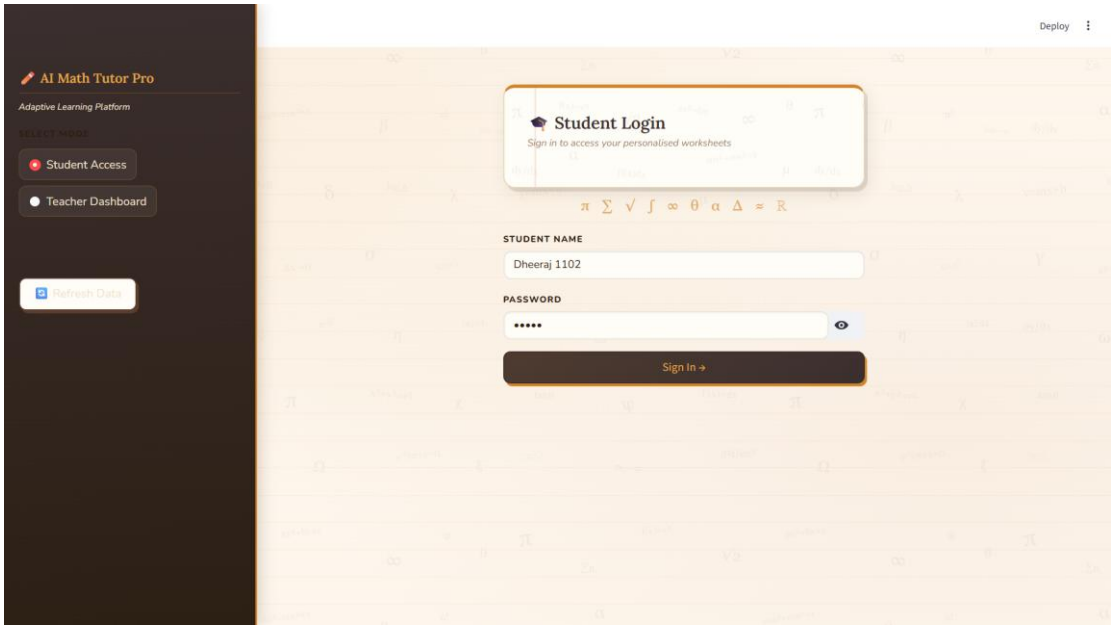

Fig. 4.3: Student Access

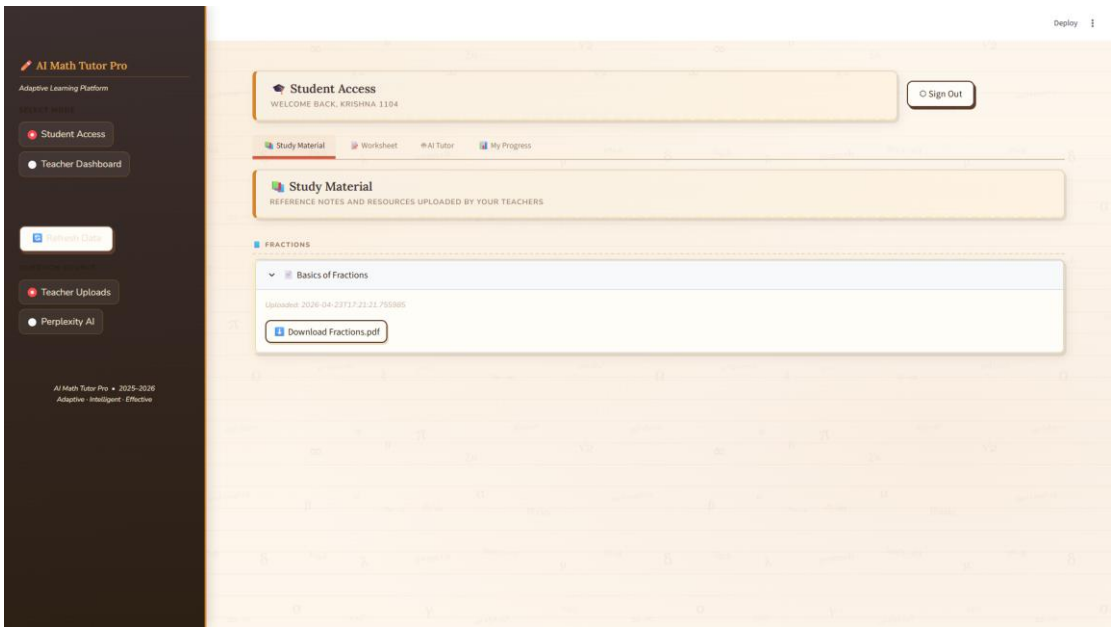

Fig. 4.4: Student Access: Study material - Fractions

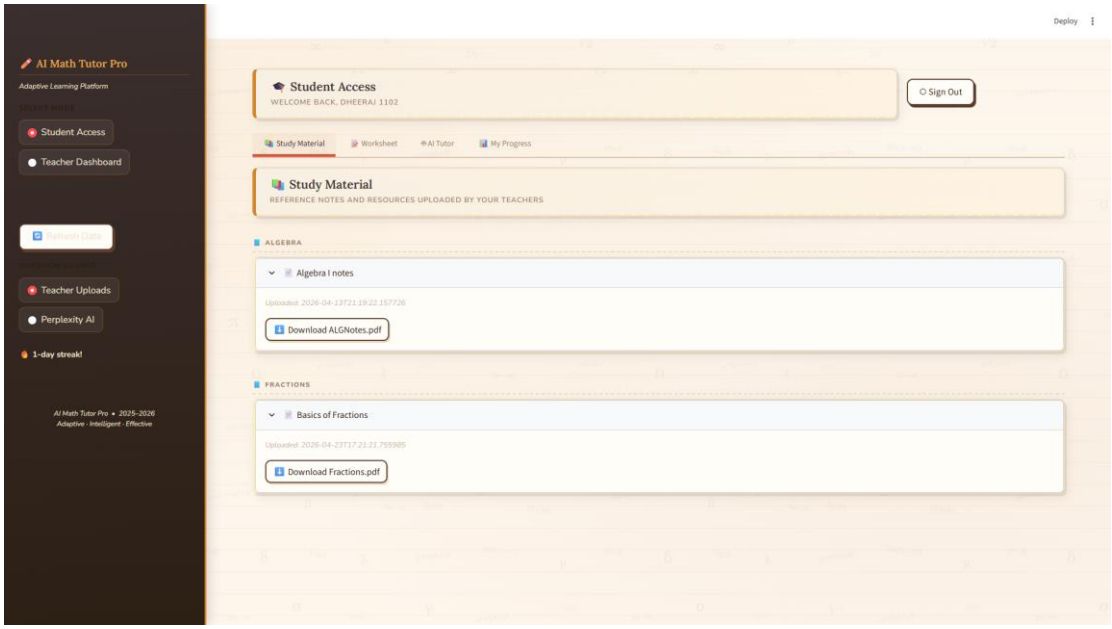

Fig. 4.5: Student Access: Study material – Algebra & Fractions

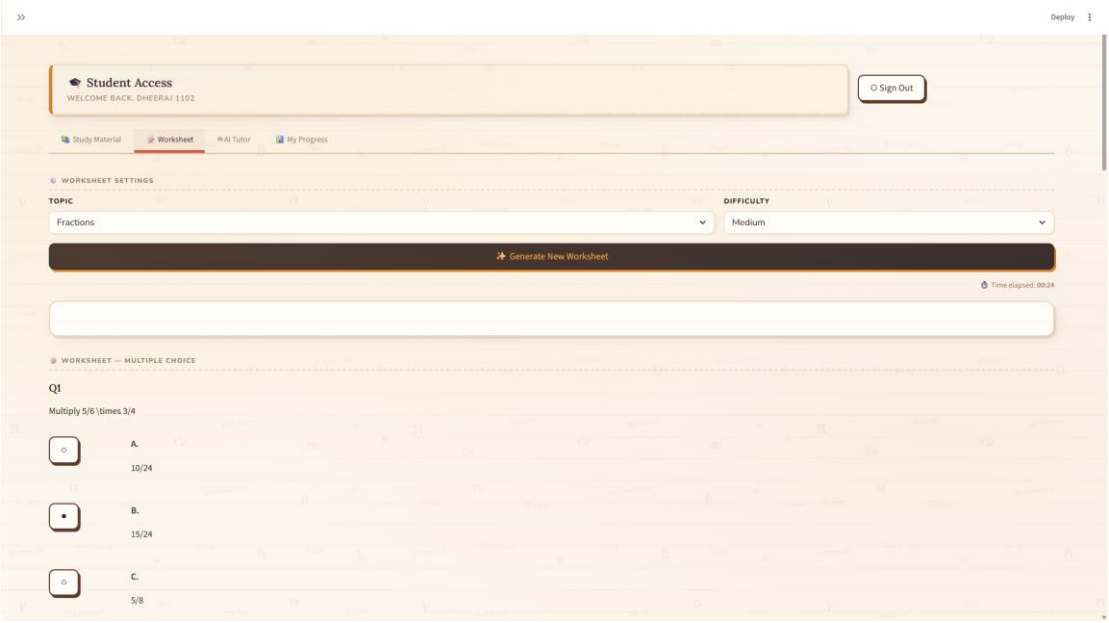

Fig. 4.6: Student Access: Worksheet Generation

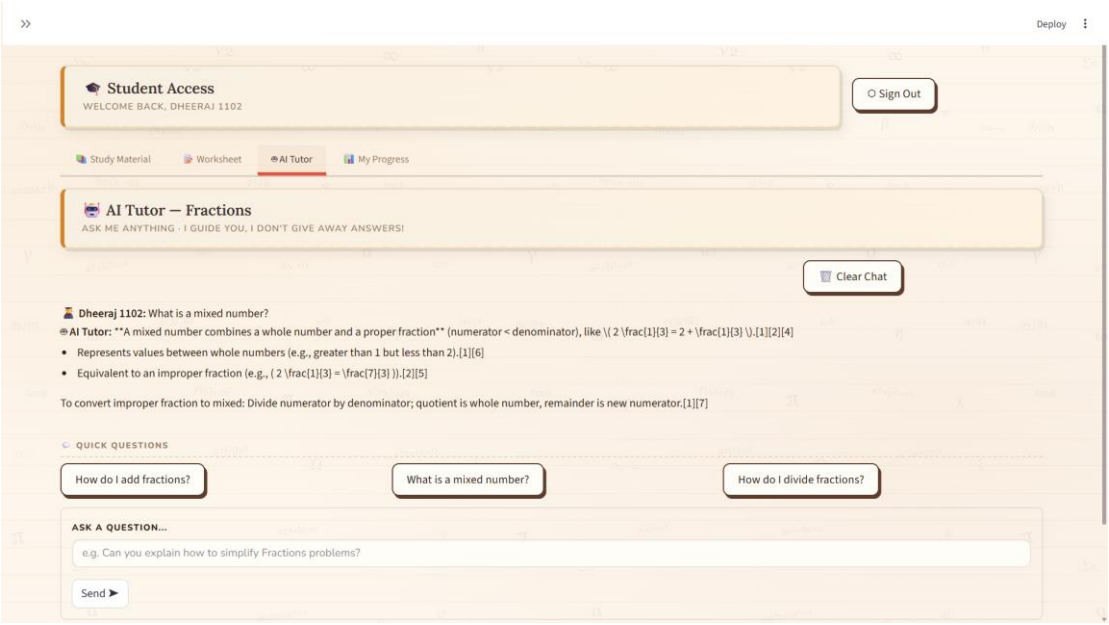

Fig. 4.7: Student Access: AI Tutor

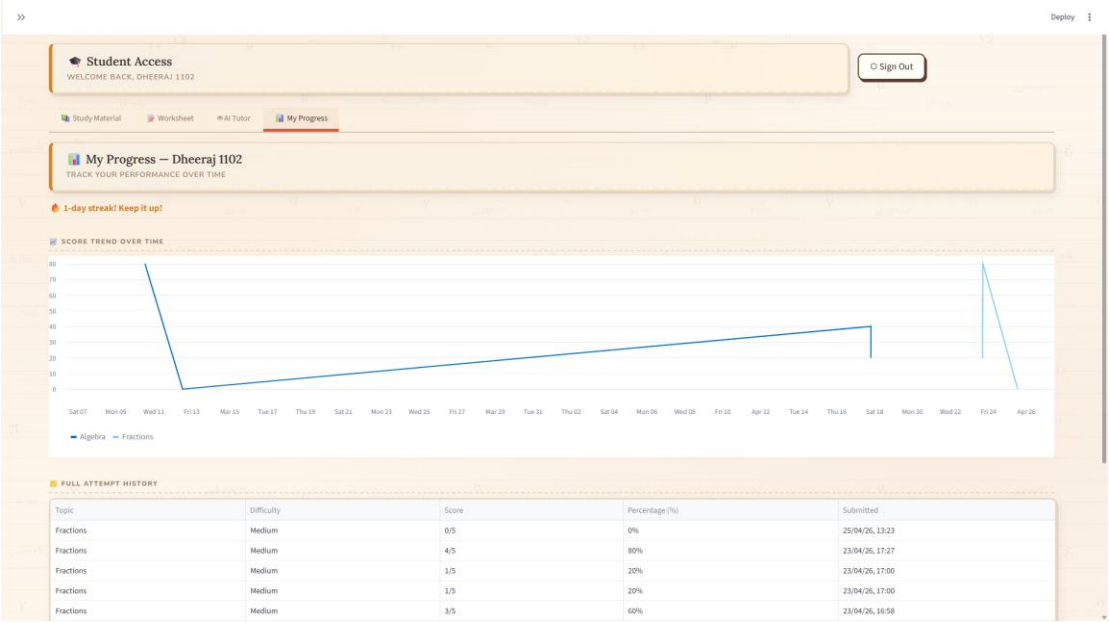

Fig. 4.8: Student Access: Student progress Charts

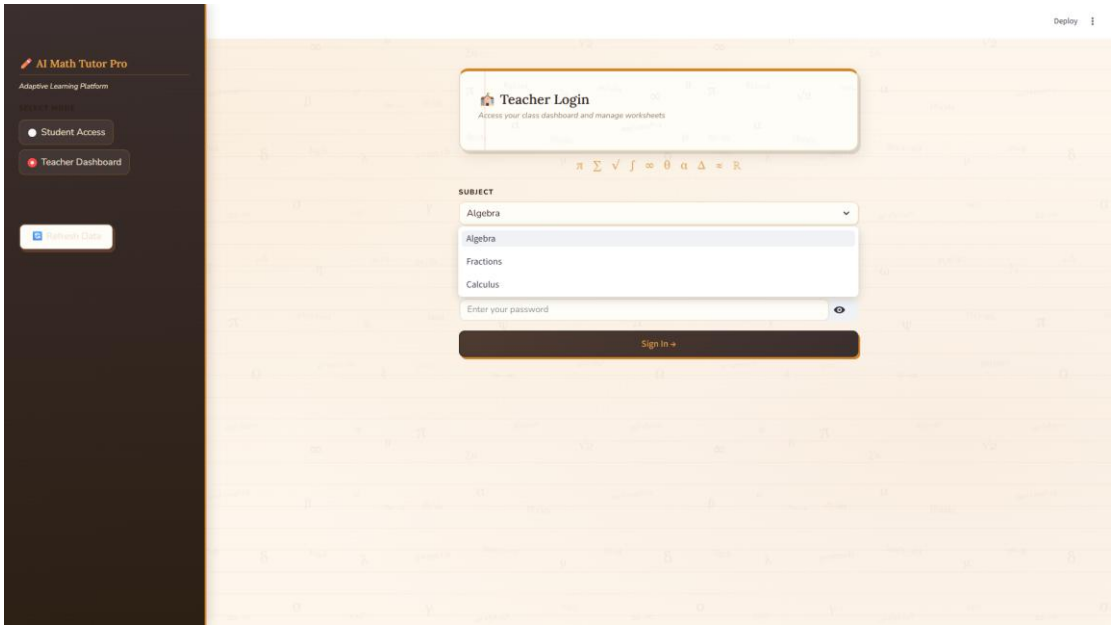

Fig. 4.9: Teacher Login: Scroll down to select subject

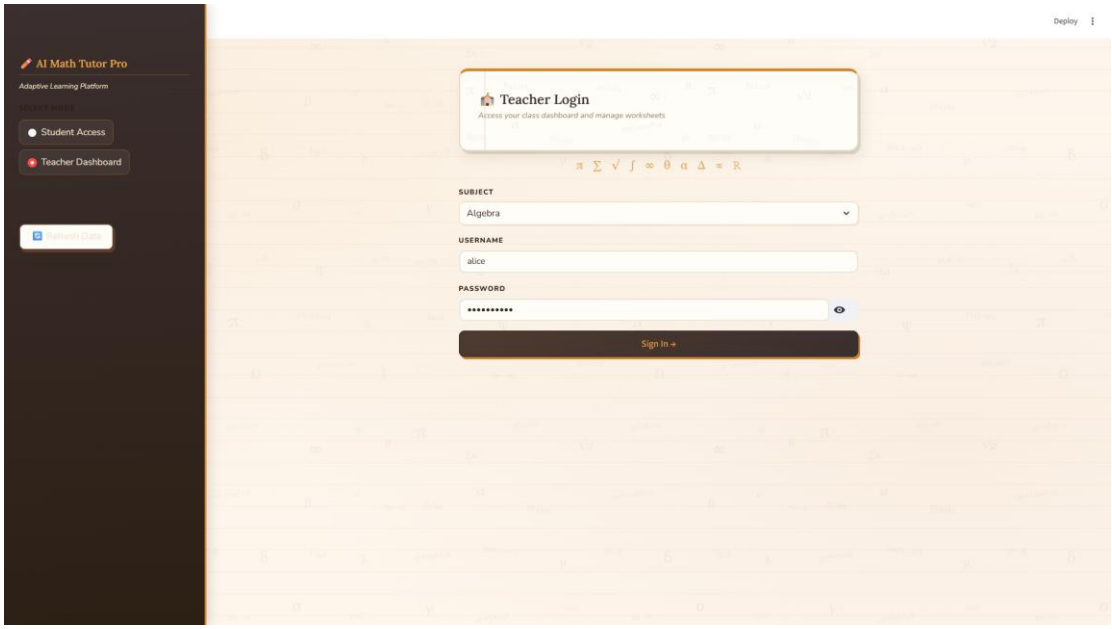

Fig. 4.10: Teacher Login with username and password

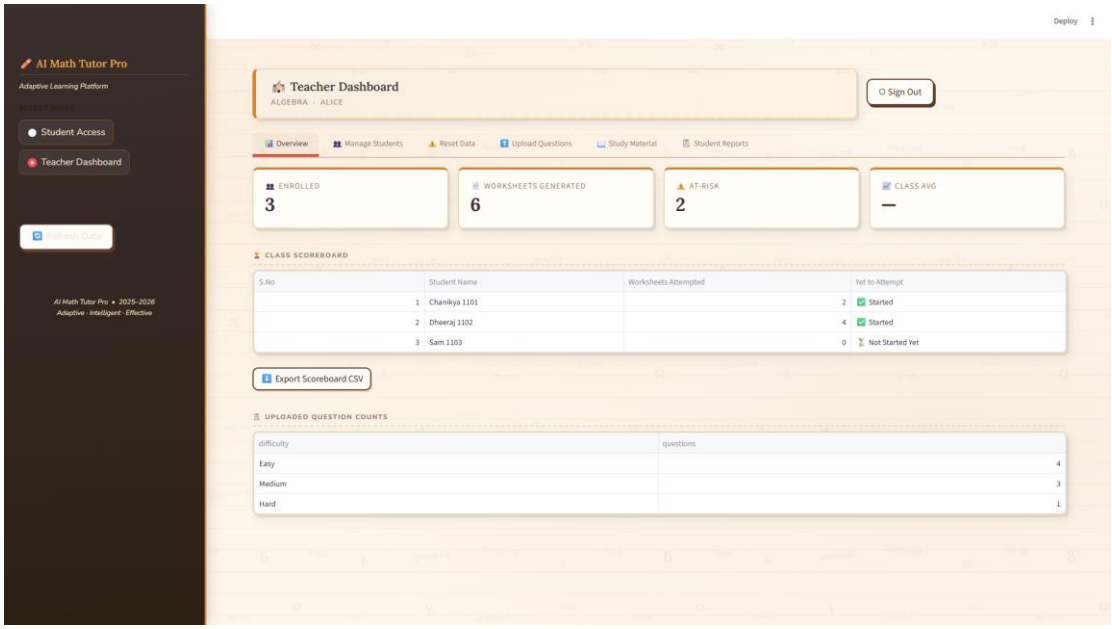

Fig. 4.11: Teacher Dashboard: Overview

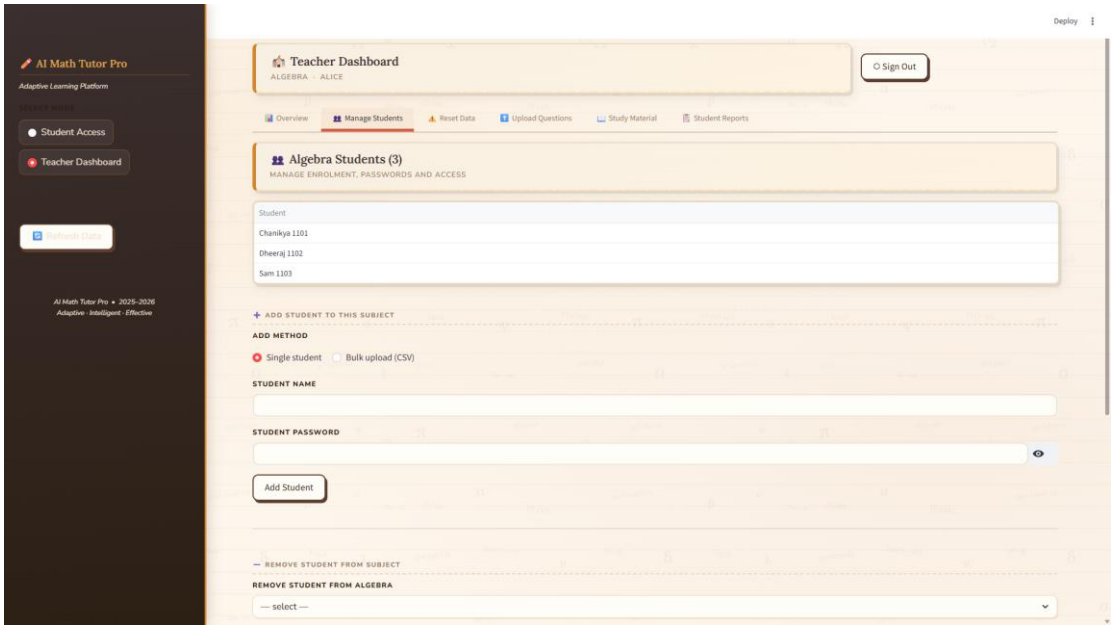

Fig. 4.12: Teacher Dashboard: Manage students

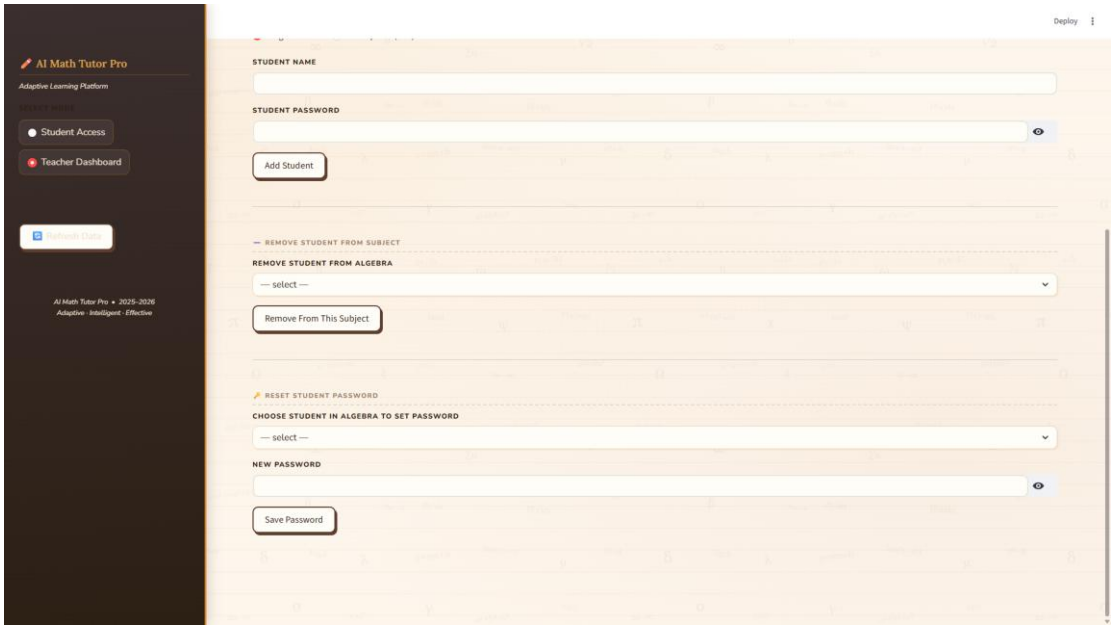

Fig. 4.13: Teacher Dashboard: Manage students – students' change password

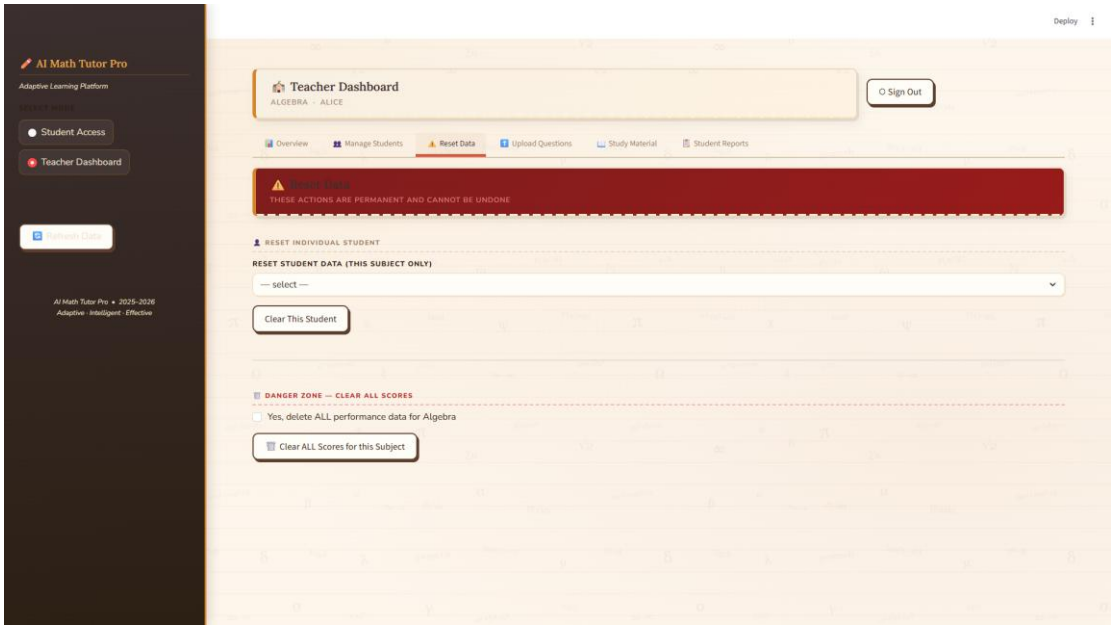

Fig. 4.14: Teacher Dashboard: Reset Data

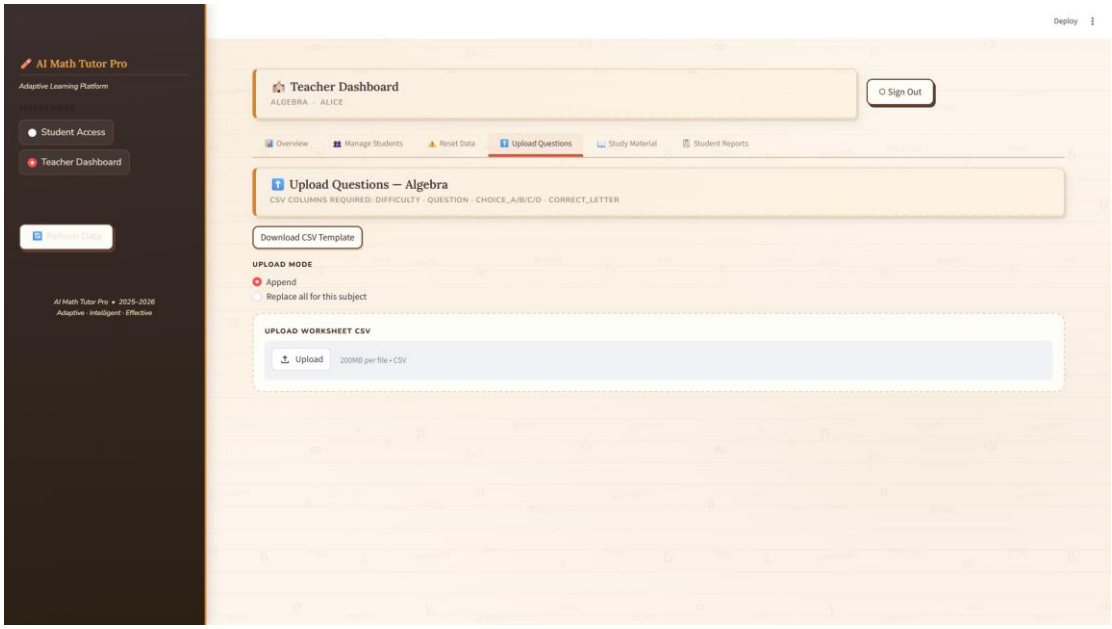

Fig. 4.15: Teacher Dashboard: Upload Questions

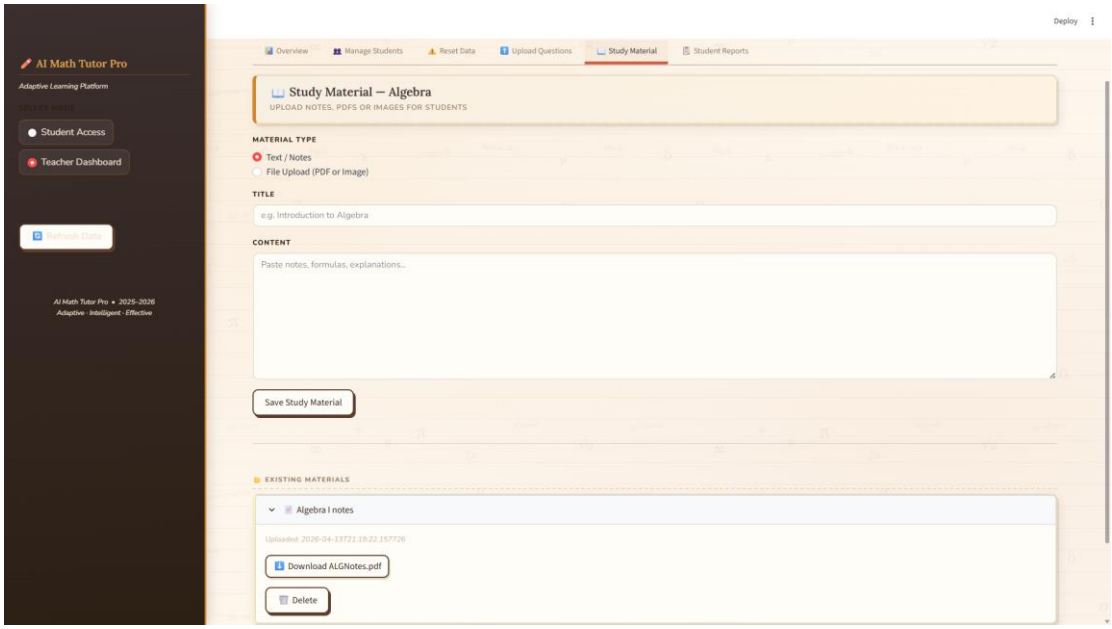

Fig. 4.16: Teacher Dashboard: Upload Study Material

CHAPTER 5
CONCLUSION

The AI Math Tutor project successfully demonstrates that an adaptive, AI-powered mathematics tutoring platform can be built and deployed using open-source, cost-free technologies without compromising on pedagogical quality or feature completeness. The system integrates the Perplexity AI sonar model for dynamic worksheet generation and Socratic tutoring, a dual-mode question system combining AI-generated and teacher-uploaded content, per-question hint delivery, adaptive difficulty progression, and a comprehensive teacher analytics dashboard — all within a locally deployable Python and Streamlit application backed by SQLite. All test cases passed successfully, validating the correctness of core functions including authentication, worksheet generation, scoring, hint delivery, streak tracking, and at-risk student identification. The result comparison confirms that AI Math Tutor outperforms both static worksheet distribution and basic online quiz platforms across every evaluated dimension.

Future development of AI Math Tutor will focus on several priority extensions that build directly on the current implementation. Cloud-based deployment using platforms such as Streamlit Community Cloud or Heroku will enable multi-school access without requiring local installation, significantly expanding the system's reach. Integration with national curriculum standards — such as the CBSE and State Board syllabi — will ensure that AI-generated questions align with formal educational requirements rather than generic mathematical topics. Voice input and text-to-speech output will improve accessibility for students with reading difficulties. A gamified reward system featuring badges, leaderboards, and milestone celebrations will strengthen student motivation for consistent daily practice. These extensions will evolve AI Math Tutor from a locally deployed classroom tool into a scalable, production-ready adaptive learning platform suitable for deployment across educational institutions of varying resource levels.

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